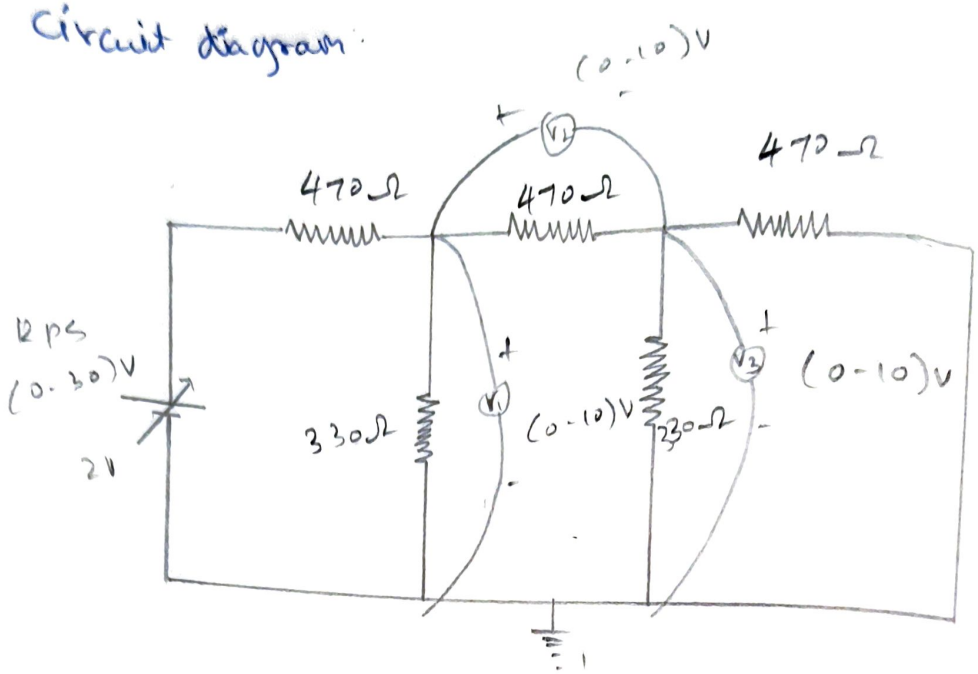
Circuit diagram:



Exp. No: 3
Date : 06/02/2026

Determination of voltage in circuit using Nodal analysis

Aim:

* To determine the voltage in the circuit using nodal analysis both theoretically and practically for a given DC circuit.

Apparatus Required:

Sl. no	Apparatus	Specification	Quantity
1.	Regulated power supply (RPS)	(0-30V)	1
2.	multimeter	–	1
4	Resistors	470Ω, 330Ω	3, 2
5.	Bread board	–	1

Theory:

* In electric circuits analysis, nodal analysis - node - voltage analysis, or the branch current method is a method of determining the voltage (potential difference) between "node" in an electrical circuit in terms of the branch currents

Tabular column:

Parameters	theoretical	Practical
V_A	0.63	0.66
V_B	0.18	0.19
$V_A - V_B$	0.45	0.47

calculations.

nodal Analysis:

nodeAl

By KCL:

$$\frac{V_A - 2}{470} + \frac{V_A}{330} + \frac{V_A - V_B}{470} = 0$$

$$\frac{33V_A - 66 + 47V_A + 33V_A - 33V_B}{15570} = 0$$

$$113V_A - 33V_B = 66 \quad - (1)$$

At Node B

By KCL:

$$\frac{V_B - V_A}{470} + \frac{V_B}{330} + \frac{V_B}{470} = 0$$

$$33V_B - 33V_A + 47V_B + 33V_B = 0$$

Procedure:

* Give connections as per the circuit diagram.
* Switch ON the supply, vary the RPS and set a particular input voltage.
* Note down the readings of ammeters and voltmeters and tabulate them.
* Reduce the RPS to its minimum value and switch OFF the supply.

Viva Voce:

* Define node
* Is nodal analysis is applicable to both dc and ac supply?
* How do we calculate branch current from the node voltages

S.No	Description	Mark allotted	Mark obtained.
1.	Experimental setup/ connection	20	20
2.	Observation / Readings	20	20
3.	calculation / Results	20	20
4.	viva	20	17
5.	Record	20	
			18/95

Result:

* Thus, the Nodal analysis is verified practically and theoretically. The resultant voltages for ——→ supply are:

a, the voltage V_1 is 0.63 V.

b, the voltage V_2 is 0.18 V.

c, the voltage V_3 is 0.45 V.